

Chapter 12

Electricity

1. The SI unit of electric current is:

- A) Volt
- B) Ohm
- C) Ampere
- D) Coulomb

Answer: C) Ampere

2. The device used to measure electric current is:

- A) Voltmeter
- B) Ammeter
- C) Galvanometer
- D) Rheostat

Answer: B) Ammeter

3. The instrument used to measure potential difference is:

- A) Ammeter
- B) Voltmeter
- C) Rheostat
- D) Galvanometer

Answer: B) Voltmeter

4. The SI unit of potential difference is:

- A) Volt
- B) Joule
- C) Ampere
- D) Ohm

Answer: A) Volt

5. The flow of electric charge per unit time is called:

- A) Voltage
- B) Resistance
- C) Current
- D) Power

Answer: C) Current

6. Ohm's law states that:

- A) $V = IR$
- B) $I = VR$
- C) $R = VI$
- D) $V = I/R$

Answer: A) $V = IR$

7. The SI unit of resistance is:

- A) Ohm
- B) Volt
- C) Ampere
- D) Watt

Answer: A) Ohm

8. The resistance of a conductor depends on:

- A) Its length
- B) Its area of cross-section
- C) The nature of the material
- D) All of these

Answer: D) All of these

9. If the length of a wire is doubled, its resistance becomes:

- A) Half
- B) Double
- C) Four times
- D) Same

Answer: B) Double

10. If the area of cross-section of a wire is doubled, its resistance becomes:

- A) Half
- B) Double
- C) Four times
- D) Same

Answer: A) Half

11. The reciprocal of resistance is called:

- A) Impedance
- B) Conductance
- C) Inductance

D) Capacitance

Answer: B) Conductance

12. The SI unit of electric charge is:

A) Ampere

B) Coulomb

C) Ohm

D) Volt

Answer: B) Coulomb

13. One coulomb of charge is equal to the charge contained in:

A) 6.25×10^{18} electrons

B) 1 electron

C) 6.25×10^6 electrons

D) 6.25×10^{12} electrons

Answer: A) 6.25×10^{18} electrons

14. In a circuit, if potential difference is 12 V and resistance is 3 Ω , then current is:

A) 36 A

B) 9 A

C) 4 A

D) 0.25 A

Answer: C) 4 A

15. In a series circuit, the total resistance is:

A) Sum of individual resistances

B) Product of resistances

C) Reciprocal of resistances

D) None

Answer: A) Sum of individual resistances

16. In a parallel circuit, the total resistance is:

A) Sum of individual resistances

B) Greater than the largest resistance

C) Less than the smallest resistance

D) None

Answer: C) Less than the smallest resistance

17. The heating effect of current is used in:

- A) Electric bulb
- B) Electric heater
- C) Electric iron
- D) All of these

Answer: D) All of these

18. The power of an electric appliance is given by:

- A) $P = VI$
- B) $P = I^2R$
- C) $P = V^2/R$
- D) All of these

Answer: D) All of these

19. The SI unit of electric power is:

- A) Joule
- B) Watt
- C) Ampere
- D) Volt

Answer: B) Watt

20. 1 kilowatt-hour is equal to:

- A) 1000 J
- B) 3.6×10^6 J
- C) 360 J
- D) 36×10^3 J

Answer: B) 3.6×10^6 J

21. The commercial unit of electrical energy is:

- A) Joule
- B) Watt
- C) Kilowatt-hour
- D) Volt

Answer: C) Kilowatt-hour

22. A fuse wire works on the principle of:

- A) Chemical effect of current
- B) Magnetic effect of current
- C) Heating effect of current

D) Electric discharge

Answer: C) Heating effect of current

23. The resistance of a good conductor is:

A) Very high

B) Very low

C) Zero

D) Infinite

Answer: B) Very low

24. The resistance of an insulator is:

A) Very low

B) Zero

C) Very high

D) Infinite

Answer: C) Very high

25. When electric current passes through a resistor, heat is produced due to:

A) Magnetic effect

B) Chemical effect

C) Collision of electrons with atoms

D) Electromagnetic effect

Answer: C) Collision of electrons with atoms

26. The electric bulb glows because of:

A) Chemical effect

B) Heating effect of current

C) Magnetic effect

D) None

Answer: B) Heating effect of current

27. The resistance of a wire depends on:

A) Material

B) Length

C) Area of cross-section

D) All of these

Answer: D) All of these

28. The potential difference between two points is 20 V and current is 4 A. Resistance is:

- A) 80 Ω
- B) 5 Ω
- C) 0.2 Ω
- D) 16 Ω

Answer: B) 5 Ω

29. The electrical energy consumed by a device is given by:

- A) $E = VIt$
- B) $E = IRt$
- C) $E = V^2Rt$
- D) $E = It^2$

Answer: A) $E = VIt$

30. The power of an electric appliance is 100 W and it operates for 10 hours. The energy consumed is:

- A) 1000 kWh
- B) 1 kWh
- C) 10 kWh
- D) 0.1 kWh

Answer: B) 1 kWh
